ChatGPT

Single-Cell Foundation Models for Cell Type Annotation and Perturbation Prediction

Abstract

Single-cell foundation models have rapidly evolved from expression-only self-supervised encoders into a broader family of transcriptomic, multimodal, ontology-aware, and language-aligned systems that aim to learn reusable representations of cellular state at atlas scale. Their most visible biological use cases are cell type annotation and perturbation prediction, because those tasks expose both the promise and the limits of transfer learning in single-cell biology: annotation tests whether pretrained representations organize stable semantic identity across tissues, batches, and ontologies, whereas perturbation prediction tests whether those representations capture causal, counterfactual, and context-dependent state transitions. This survey synthesizes the field through July 2026, with emphasis on model design, pretraining data, adaptation strategies, evaluation protocols, and biological realism. We argue that the literature has converged on three central lessons. First, pretraining on tens of millions of cells can improve robustness, especially when combined with ontology structure, multimodal priors, or language supervision. Second, gains are highly task- and protocol-dependent, and zero-shot claims often weaken under stronger controls, distribution shifts, and realistic benchmark design. Third, perturbation prediction remains the harder and more revealing target: many models still underperform simple baselines in some settings, yet recent evidence suggests that carefully chosen foundation representations, especially multimodal or fused ones, can materially improve generalization. The field is therefore moving away from a simplistic "larger is always better" narrative toward a more disciplined program centered on curation, causality, adaptation efficiency, and benchmark fidelity. [1]

Introduction

Single-cell transcriptomics has transformed the study of cellular heterogeneity by enabling cell-resolved measurements across development, homeostasis, disease, and experimental intervention. As data volumes expanded from small studies to atlas-scale resources, the core analytical bottlenecks shifted from generating measurements to organizing them into reusable knowledge. Two tasks became especially central. The first is **cell type annotation**, which converts sparse, noisy transcriptional profiles into semantically meaningful labels. The second is **perturbation prediction**, which asks how a cell's molecular state would change under a genetic or chemical intervention, even when that exact intervention has not been measured. These tasks are linked but not identical: annotation is primarily about stable identity under nuisance variation, whereas perturbation prediction is about conditional dynamics and counterfactual inference. Both have been catalyzed by the emergence of giant public resources such as the Human Cell Atlas, Tabula Sapiens, cross-tissue immune atlases, and CZ CELLxGENE Discover. [2]

Before foundation models, single-cell annotation and perturbation modeling developed as mostly separate literatures. Annotation relied on reference mapping, marker-based methods, hierarchical classifiers, and domain-specific deep models. Perturbation modeling built on variational autoencoders, conditional generative models, optimal transport, and graph-informed architectures. Foundation models changed the landscape by proposing a common recipe: pretrain on massive unlabeled corpora with self-supervised or weakly supervised objectives, then adapt a shared representation across downstream tasks. In single-cell biology this idea has produced what recent reviews call large cellular models or single-cell foundation models, spanning transcriptome-only transformers, rank-based encoders, asymmetric sparse architectures, ontology-regularized models, graph-based systems, cross-species models, multimodal text-cell encoders, and increasingly general-purpose benchmarking frameworks. 3

The initial wave of highly visible models included Geneformer, scGPT, scFoundation, xTrimoGene, scBERT, tGPT, UCE, CellPLM, and scMulan, followed by SCimilarity, LangCell, CellWhisperer, CellFM, scPRINT, Cell-GraphCompass, EpiAgent, scCello, AIDO.Cell, RegFormer, CellVQ, and related systems. These models differ substantially in what is tokenized, what invariances are assumed, and how biological priors are injected. Some treat a cell as an ordered sequence of genes; some use binned or masked expression values; some model cells as graphs; some align cells to natural language or protein embeddings; some explicitly exploit cell ontology or read-depth information. Their downstream claims also vary. Some emphasize fine-tuned annotation, some emphasize zero-shot cell identity understanding, some emphasize retrieval, and some target perturbation extrapolation or drug-response inference. 4

At the same time, the field has matured enough to generate a serious evaluation literature. Critical studies have shown that many apparent gains diminish under stronger baselines, realistic batch shifts, or stricter zero-shot settings; that metric choice can distort conclusions about embedding quality; and that perturbation prediction is especially sensitive to benchmark design and data leakage. Large benchmark efforts in 2025 and 2026 have therefore reframed the field: the main question is no longer whether single-cell foundation models are interesting, but when, why, and under what assumptions they outperform narrower methods. This survey adopts exactly that framing. It organizes the literature around method families and research objectives, then evaluates how those families bear on cell type annotation and perturbation prediction in practice. 5

Methods

A useful taxonomy begins with what the model treats as primitive. Most transcriptome foundation models operate on gene identities, gene ranks, or discretized expression values. A smaller but increasingly important family uses cells, tissues, or neighborhoods as tokens, allowing cell–cell context to shape the learned representation. A third family aligns transcriptomes to external modalities such as text, protein language embeddings, chromatin accessibility, or spatial context. Across these families, the key design choices are tokenization, ordering, sparsity handling, objective function, and adaptation interface. This yields a methodological continuum rather than a single paradigm: masked modeling, causal generation, contrastive alignment, metric learning, graph pretraining, ontology-regularized pretraining, and multimodal co-embedding now coexist in the literature. Representative works are surveyed in [1-34]. 6

Transcriptome-only pretraining remains the dominant backbone for cell type annotation. Geneformer [4] uses rank-value encoding to emphasize relative expression and network context; scGPT [5] extends generative pretraining to single-cell multi-omics; scFoundation [6] and xTrimoGene [7] exploit asymmetric or sparse architectures for scalability; scBERT [8] and tGPT [9] adapt transformer language-modeling ideas more directly to transcriptomics; UCE [10] introduces protein-informed gene representations and cross-species ambitions; CellPLM [11] reframes the problem by treating cells as tokens and tissues as sentences; and scTab [14] demonstrates that even non-transformer tabular architectures can scale effectively when annotation is the target. SCimilarity [16] further shows that metric-learning style foundation models can enable atlas-scale retrieval and cell search, broadening the notion of “annotation” from label prediction to similarity-based contextualization. Collectively, these works suggest that the best transcriptomic pretraining strategy depends less on architectural fashion than on alignment between tokenization, data sparsity, and the downstream transfer protocol. 7

Biological structure is increasingly injected explicitly rather than left to emerge implicitly. GeneCompass [15] incorporates prior regulatory knowledge across species; scCello [26] uses cell ontology coherence and ontology-alignment losses; Cell-GraphCompass [19] and RegFormer [30] introduce graph or regulatory priors into the representation of each cell; EpiAgent [20] extends the foundation-model recipe from scRNA-seq to scATAC-seq; and CellVQ [29] adds a discrete cell-code interface to improve interpretability. This trajectory reflects a broader shift in the field. Early work often hoped that large-scale self-supervision alone would recover biologically meaningful abstractions. Recent work instead treats priors, ontologies, and graph structure as necessary inductive bias for robust generalization, especially when the evaluation moves beyond easy in-distribution splits. These models are especially relevant to annotation because hierarchical relations among cell types are real biological structure, not merely a bookkeeping convention. 8

Multimodal and language-aligned models try to close the semantic gap between expression and biological meaning. LangCell [12] is a landmark in this direction because it aligns transcriptomes with natural-language descriptions of cell identity, enabling zero-shot and few-shot cell identity understanding. CellWhisperer [21] pushes further by learning a multimodal interface for conversational exploration of single-cell data. Related works such as sciLaMA [22], scMMGPT [23], Cell2Sentence [24], GenePT [25], LICT [33], scAgent [67], CellMaster [68], and scMPT [69] use large language models either as teachers, semantic priors, or inference-time reasoning modules rather than as direct replacements for expression-native encoders. The methodological point is subtle but important: natural language helps most when it supplies ontology, context, and descriptive semantics that are not identifiable from raw counts alone. It helps less when the task is dominated by distributional geometry already recoverable from expression. For annotation, this semantic supervision can be powerful when labels are sparse or hierarchical. For perturbation prediction, its value is more indirect, often entering through better cell-state descriptions, gene semantics, or drug-response metadata. 9

Adaptation for cell type annotation now follows four main patterns. The first is linear probing or shallow classification on frozen pretrained embeddings, often used to test whether identity is already linearly separable. The second is full or parameter-efficient fine-tuning, as in LoRA-style adaptation, when domain shift is substantial. The third is reference mapping or retrieval, where the model embeds query cells and transfers labels from nearest atlas neighbors. The fourth is text- or ontology-mediated prediction, where candidate labels live in a structured semantic space. Pre-foundation annotation methods [35-66] remain highly relevant because they reveal what the foundation-model era is actually trying to improve upon: robustness to platform shift, hierarchical consistency, rare-cell recall, novelty detection, and standardization of labels. Modern foundation approaches improve these dimensions mainly when the pretraining corpus is broad and when the downstream protocol respects ontology and batch structure. However, strong domain-specific tools such as scmap, SingleR, Garnett, scPred, ACTINN, CHETAH, scClassify, SingleCellNet, CellAssign, Cell BLAST, SciBet, scANVI, scArches, ItClust, CellO, JIND, scGCN, scDeepSort, scAnnotate, and Census remain competitive or superior in many realistic settings. 10

Perturbation prediction is methodologically richer and scientifically harder. The classical lineage runs from scGen [70] and trVAE [71], which learn latent perturbation vectors or conditional decoders, through CPA [72] and MultiCPA [73], which factorize cell state, perturbation, dosage, and covariate effects, to CellOT [74], which models perturbational transitions through neural optimal transport. Later models such as scVIDR [75], scPreGAN [76], GEARS [77], biolord [78], scDisInFact [79], and scPRAM [80] enrich this toolkit with dosage reasoning, adversarial generation, graph priors, disentangled latent variables, and attention-based prediction. The foundation-model connection originally came from the hope that large pretrained cell encoders would supply better priors for these conditional generators. That idea is plausible because perturbation response depends partly on the baseline cellular manifold, but the empirical record is mixed: the ability to represent baseline state does not automatically translate into reliable counterfactual prediction under unseen interventions. 11

Recent perturbation-focused work has therefore emphasized benchmark realism, adaptation protocols, and multimodal representation choice. Benchmarking algorithms for generalizable single-cell perturbation response prediction [82], the OP3 benchmark [84], PertEval-scFM [85], scDrugMap [88], scArchon [95], and other evaluations have highlighted the importance of split design across unseen cell contexts, perturbations, and combination regimes. A striking result from 2025 was that several federal-wave foundation models and deep perturbation predictors failed to beat simple linear baselines on heterogeneous genetic perturbation data [83]. Yet this is not the final word. More recent analyses indicate that some foundation representations, especially when efficiently fine-tuned [86] or fused across modalities [89, 90], improve perturbation response modeling for both chemical and genetic settings. The emerging consensus is that perturbation prediction rewards the right representation more than the largest representation: interactome-, ontology-, or multimodal-informed features can matter more than sheer parameter count or corpus size. 12

Evaluation practice has become a research area in its own right. Open Problems in Single-Cell Analysis [103], scEval [104], BioLLM [105], benchmark studies on transcriptomic foundation models [106], assessments of class imbalance [107], work on metric failure modes [109], and critiques of zero-shot transfer [101, 110] have shown that the same model can appear excellent or mediocre depending on ontology granularity, batch composition, negative-class construction, and whether novelty or rare-cell scenarios are included. Two consequences follow. First, cell type annotation should be reported at multiple resolutions, with uncertainty and reject options when possible. Second, perturbation prediction should be evaluated not only on whole-transcriptome similarity but also on differential-expression fidelity, pathway-level effects, combination interactions, and practical extrapolation regimes. In other words, “foundation-model performance” is not a scalar property; it is a function of the biological question, the nuisance factors, and the benchmark’s hidden assumptions. 13

Challenges and Prospects

A first major challenge is **data curation and ontology consistency**. Atlas-scale pretraining depends not only on cell count but also on harmonization of labels, tissues, disease states, perturbation metadata, and gene vocabularies. The literature increasingly shows that ontology structure is a major resource for annotation, yet the underlying corpus remains heterogeneous and partially inconsistent even in premier resources. This is not a minor engineering detail. If labels are noisy, tissue-specific, or non-comparable across studies, then a pretrained model may learn shortcuts about source study, chemistry, or protocol rather than cell identity. The strongest recent annotation systems therefore combine scale with explicit ontology regularization, curated subsets, or hierarchical loss design rather than relying on raw aggregation alone. References [14-16], [26], [33], [55], [58], [63], [112-116], [137-139] are especially informative on this point. 14

A second challenge is **distribution shift and the fragility of zero-shot claims**. Zero-shot cell identity understanding is appealing because it promises atlas-scale reuse without task-specific retraining. But several studies now show that apparent zero-shot utility can erode under batch shift, class imbalance, real-world integration settings, or dynamics-oriented evaluation. More generally, robust performance on random train–test splits within a curated dataset is not the same as robust performance across tissues, donors, species, diseases, or time. For annotation, this means that frozen embeddings alone are unlikely to be universally sufficient. For perturbation prediction, it means that a good baseline-state embedding is still not a proof of causal extrapolation. References [99-111], and especially [101], [106], [107], [109], [110], argue for treating universal embeddings as a target that remains aspirational rather than solved. 15

A third challenge is **causal faithfulness in perturbation prediction**. Many models can interpolate within familiar regimes or reconstruct average transcriptional shifts, but truly useful perturbation models must extrapolate to unseen combinations, dosages, cell states, and mechanism classes. The 2025 Nature Methods benchmark and related critiques make clear that this goal is still only partially achieved. The most promising direction is not to abandon foundation models, but to integrate them with causal structure: gene regulatory priors, interaction networks, dosage-aware mechanisms, uncertainty quantification, and richer perturbation corpora such as Perturb-seq and multiplexed chemical transcriptomics. In this respect, perturbation prediction may become the decisive proving ground for "virtual cell" ambitions, because it forces models to represent state transitions rather than merely state similarity. References [70-95], [125-132], and [135] provide the conceptual bridge from current predictive systems to more mechanistically grounded future ones. 16

A fourth challenge, which is also the clearest opportunity, is **compositional multimodality**. The next generation of single-cell foundation models is unlikely to remain transcriptome-only. Language-aligned models expose semantic priors; epigenomic models capture regulatory potential; multi-omics integration models measure where foundation methods still lag specialist tools; and recent perspectives argue for compositional systems that can combine transcriptomics, chromatin, protein abundance, spatial localization, microscopy, and text. For cell type annotation, this should improve semantic specificity, uncertainty estimation, and rare-state recognition. For perturbation prediction, it should improve mechanism awareness and generalization under sparse supervision. The frontier is therefore not merely bigger transformers, but a modular ecosystem linking pretrained cell representations with domain knowledge, retrieval, structured ontologies, and experimental design loops. References [20-25], [29-34], [104-111], [119-140] suggest that such a transition is already underway. 17

Conclusion

Single-cell foundation models have established themselves as a durable research direction, but not because they have already solved cell biology. Their significance is that they offer a common representation-learning framework through which annotation, retrieval, integration, and perturbation modeling can be studied in relation rather than isolation. For cell type annotation, they are most compelling when they combine scale with ontology, retrieval, or multimodal semantics; for perturbation prediction, they are most promising when they are embedded inside conditional, graph-informed, or multimodal forecasting pipelines rather than used as generic embeddings by default. The empirical literature now supports a balanced conclusion. Foundation models can yield meaningful gains, but those gains are conditional on curation quality, benchmark realism, and adaptation design. The field's next advances will likely come from better data and better inductive bias as much as from larger parameter counts. In that sense, the transition from "single-cell large models" to scientifically useful virtual cell systems remains a work in progress, but it is no longer speculative: it is an increasingly testable, benchmarkable program. 18

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